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NEET · PHYSICS · CHAPTER-WISE PYQS

Capacitor

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	39 questions
Answer key	at the end

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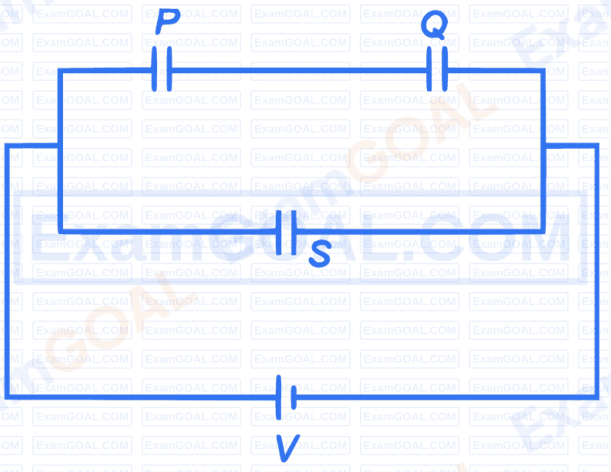
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MCQ (Single Correct Answer) 39 questions

Q1. RE-NEET 2026

Open solution

Three identical capacitors, P, Q and S, each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\frac{U_P}{U_T}$ is:



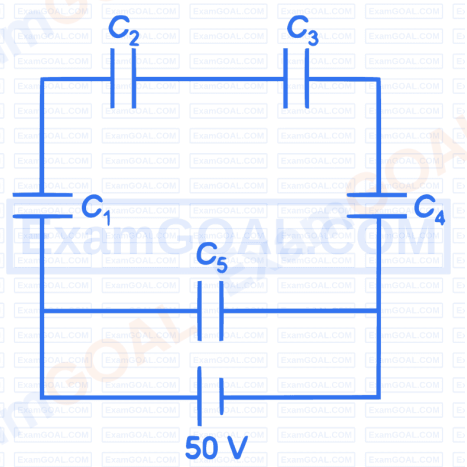
(A) $1/6$ (B) $2/3$ (C) $1/3$ (D) $1/2$

Q2. NEET 2026

Open solution

Five capacitors of capacitances

$C_1 = C_2 = C_3 = C_4 = 10\mu\,\mathrm{F}$ and $C_5 = 2.5\mu\,\mathrm{F}$ are connected as shown, along with a battery of 50 V .



The equivalent capacitance and the charges on each capacitor respectively are:

(A) $5\mu\,\mathrm{F}, 125\mu\mathrm{C}$ on C_1 to C_4 and $25\mu\mathrm{C}$ on C_5

(B) $5\mu\,\mathrm{F}, 125\mu\mathrm{C}$ on all capacitors

(C) $5\mu\,\mathrm{F}, 250\mu\mathrm{C}$ on all capacitors

(D) $4\mu\,\mathrm{F}, 250\mu\mathrm{C}$ on C_1 to C_4 and $125\mu\mathrm{C}$ on C_5

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Q3. NEET 2026

Consider two uncharged capacitors of equal capacitance 200 pF . One of them is charged by a 100 V supply and disconnected. Now this capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in the process is:

Open solution

(A) 0.5×10^{-6} J (B) 1.0 J (C) 1.0×10^{-6} J (D) 0.5 J

Q4. NEET 2025

The plates of a parallel plate capacitor are separated by d. Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates. If $K_1 = 1.25K_2$, the value of K_1 is:

Open solution

(A) 1.60 (B) 1.33 (C) 2.66 (D) 2.33

Q5. NEET 2024 (Re-Examination)

A $12\,\mathrm{pF}$ capacitor is connected to a $50\,\mathrm{V}$ battery, the electrostatic energy stored in the capacitor in nJ is

Open solution

(A) 15 (B) 7.5 (C) 0.3 (D) 150

Q6. NEET 2024 (Re-Examination)

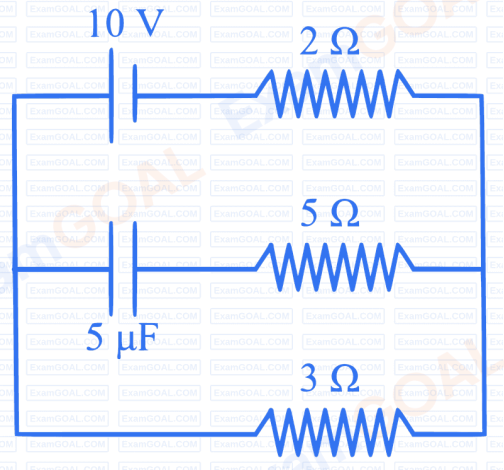
The capacitance of a capacitor with charge q and a potential difference V depends on

Open solution

(A) both q and V (B) the geometry of the capacitor
(C) q only (D) V only

Q7. NEET 2024 (Re-Examination)

The steady state current in the circuit shown below is :



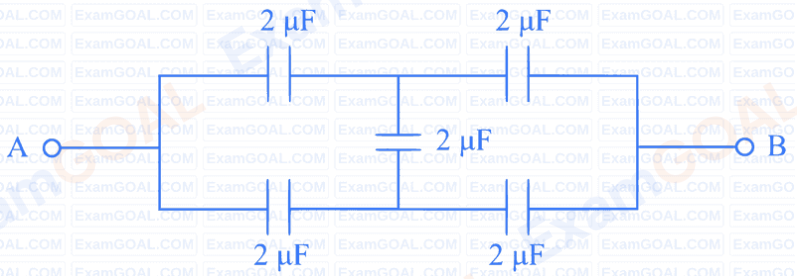
Open solution

(A) 0.67 A (B) 1.5 A (C) 2 A (D) 1 A

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Q8. NEET 2024

In the following circuit, the equivalent capacitance between terminal A and terminal B is :



Open solution

(A) 2 μF (B) 1 μF (C) 0.5 μF (D) 4 μF

Q9. NEET 2024

If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then

A. the charge stored in it, increases.

B. the energy stored in it, decreases.

C. its capacitance increases.

D. the ratio of charge to its potential remains the same.

E. the product of charge and voltage increases.

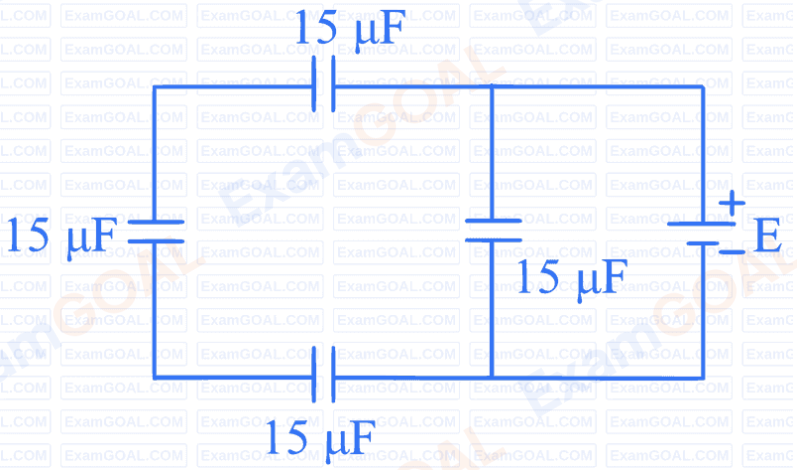
Choose the most appropriate answer from the options given below:

(A) A, B and E only (B) A, C and E only (C) B, D and E only (D) A, B and C only

Open solution

Q10. NEET 2023 Manipur

The equivalent capacitance of the arrangement shown in figure is:



(A) 30 μF (B) 15 μF (C) 25 μF (D) 20 μF

Open solution

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Q11. NEET 2023 Manipur

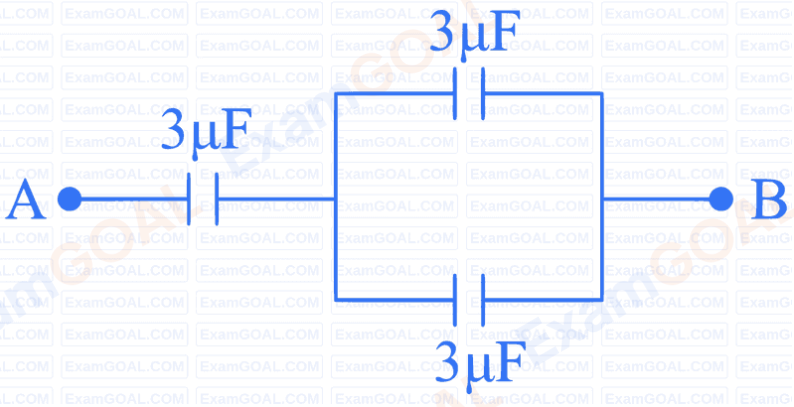
To produce an instantaneous displacement current of $2\ \mathrm{mA}$ in the space between the parallel plates of a capacitor of capacitance $4\ \mu\mathrm{F}$, the rate of change of applied variable potential difference $\left(\frac{\mathrm{dV}}{\mathrm{dt}}\right)$ must be :-

Open solution

(A) $800\ \mathrm{V/s}$ (B) $500\ \mathrm{V/s}$ (C) $200\ \mathrm{V/s}$ (D) $400\ \mathrm{V/s}$

Q12. NEET 2023

The equivalent capacitance of the system shown in the following circuit is:



Open solution

(A) 3μF (B) 6μF (C) 9μF (D) 2μF

Q13. NEET 2022 Phase 2

The distance between the two plates of a parallel plate capacitor is doubled and the area of each plate is halved. If C is its initial capacitance, its final capacitance is equal to

Open solution

(A) $\frac{C}{4}$ (B) 2C (C) $\frac{C}{2}$ (D) 4C

Q14. NEET 2022 Phase 2

The effective capacitances of two capacitors are 3 μF and 16 μF, when they are connected in series and parallel respectively. The capacitance of two capacitors are :

Open solution

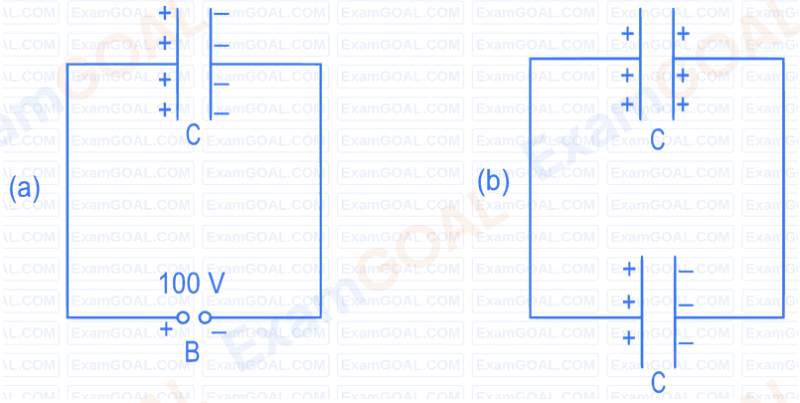
(A) 1.2 μF, 1.8 μF (B) 10 μF, 6 μF (C) 8 μF, 8 μF (D) 12 μF, 4 μF

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Q15. NEET 2022 Phase 1

Open solution

A capacitor of capacitance C = 900 pF is charged fully by 100 V battery B as shown in figure (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance C = 900 pF as shown in figure (b). The electrostatic energy stored by the system (b) is



(A) 4.5×10^{-6} J (B) 3.25×10^{-6} J (C) 2.25×10^{-6} J (D) 1.5×10^{-6} J

Q16. NEET 2021

Open solution

A parallel plate capacitor has a uniform electric field $\vec{E}$ in the space between the plates. If the distance between the plates is 'd' and the area of each plate is 'A', the energy stored in the capacitor is : (ε_0 = permittivity of free space)

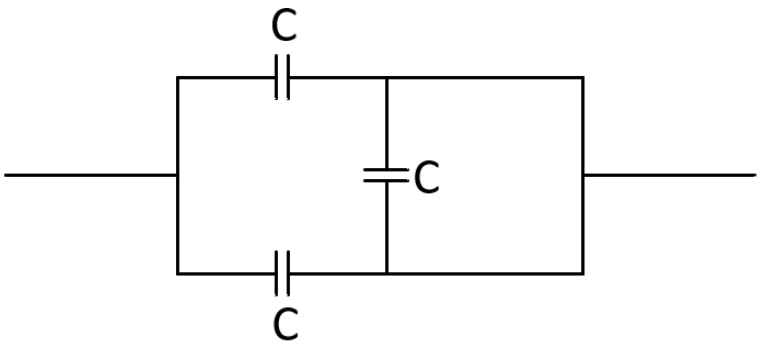
(A) $\frac{E^2 Ad}{\varepsilon_0}$ (B) $\frac{1}{2}\varepsilon_0 E^2$

(C) $\varepsilon_0 EAd$ (D) $\frac{1}{2}\varepsilon_0 E^2 Ad$

Q17. NEET 2021

Open solution

The equivalent capacitance of the combination shown in the figure is :



(A) 3C/2 (B) 3C (C) 2C (D) C/2

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Q18. NEET 2020 Phase 1

The capacitance of a parallel plate capacitor with air as medium is 6μF. With the introduction of a dielectric medium, the capacitance becomes 30 μF. The permittivity of the medium is :
$(\varepsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2})$

Open solution

(A) $1.77 \times 10^{-12} C^2 N^{-1} m^{-2}$
(B) $0.44 \times 10^{-10} C^2 N^{-1} m^{-2}$
(C) $5.00 C^2 N^{-1} m^{-2}$
(D) $0.44 \times 10^{-13} C^2 N^{-1} m^{-2}$

Q19. NEET 2019

A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3V/s. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor, would be, respectively :

Open solution

(A) zero, zero
(B) zero, 60 μA
(C) 60 μA, 60 μA
(D) 60 μA, zero

Q20. NEET 2018

The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

Open solution

(A) Independent of the distance between the plates
(B) Linearly proportional to the distance between the plates
(C) Proportional to the square root of the distance between the plates
(D) Inversely proportional to the distance between the plates

Q21. NEET 2017

A capacitor is charged by a battery. The battery is removed and another identical uncharged capacitor is connected in parallel. The total electrostatic energy of resulting system

Open solution

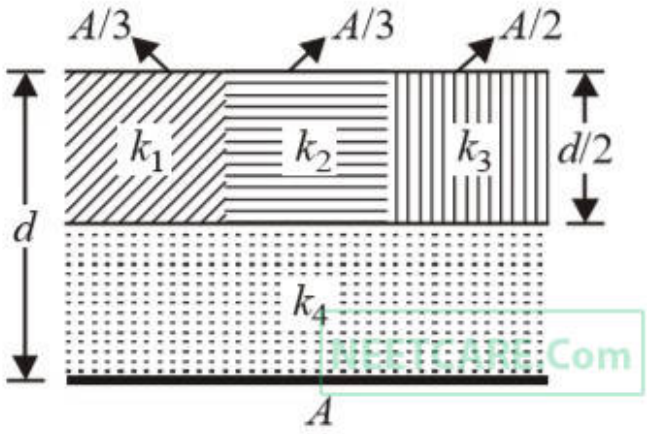
(A) decreases by a factor of 2
(B) remains the same
(C) increases by a factor of 2
(D) increases by a factor of 4

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Q22. NEET 2016 Phase 2

A parallel-plate capacitor of area A, plate separation d and capacitance C is filled with four dielectric materials having dielectric constants k_1, k_2, k_3 and k_4 as shown in the figure. If a single dielectric material is to be used to have the same capacitance C in this capacitor, then its dielectric constant k is given by

Open solution



(A) $k = k_1 + k_2 + k_3 + 3k_4$

(B) $k = \frac{2}{3}(k_1 + k_2 + k_3) + 2k_4$

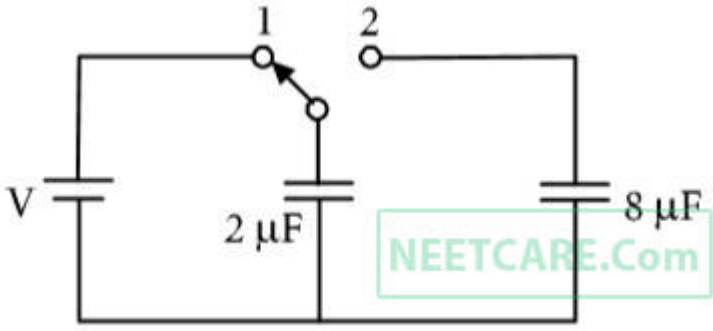
(C) $\frac{2}{k} = \frac{3}{k_1+k_2+k_3} + \frac{1}{k_4}$

(D) $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \frac{3}{2k_4}$

Q23. NEET 2016 Phase 1

A capacitor of 2 μF is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its stored energy dissipated is

Open solution



(A) 75% (B) 80% (C) 0% (D) 20%

Q24. AIPMT 2015

A parallel plate air capacitor has capacity C, distance of separation between plates is d and potential difference V is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is

Open solution

(A) $\frac{CV^2}{d}$ (B) $\frac{C^2V^2}{2d^2}$ (C) $\frac{C^2V^2}{2d}$ (D) $\frac{CV^2}{2d}$

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Q25. AIPMT 2015 Cancelled Paper

Open solution

A parallel plate air capacitor of capacitance C is connected to a cell of emf V and then disconnected from it. A dielectric slab of dieletric constant K, which can just fill the air gap of the capacitor, is now inserted in it . Which of the following is incorrect ?

(A) The change in energy stored is $\frac{1}{2}CV^2\left(\frac{1}{K}-1\right)$

(B) The charge on the capacitor is not conserved.

(C) The potential difference between the plates decreases K times.

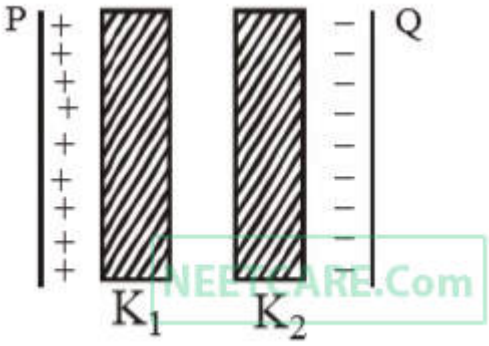
(D) The energy stored in the capaciotor decreases K times.

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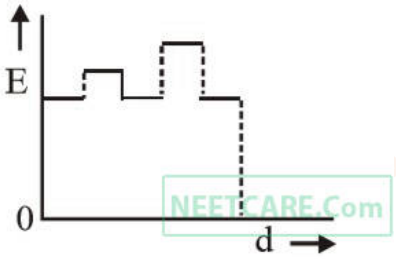
Q26. AIPMT 2014

Two thin dielectric slabs of dielectric constants K_1 and $K_2(K_1 < K_2)$ are inserted between plates of a parallel plate capacitor, as shown in the figure. The variation of electric field E between the plates with distance d as measured from plate P is correctly shown by

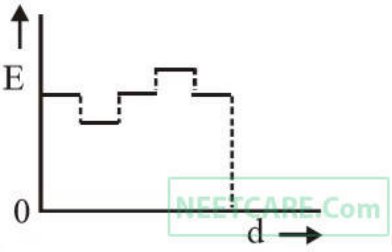
Open solution



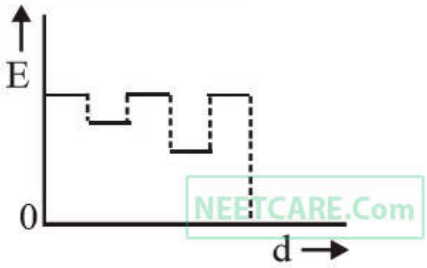
(A)



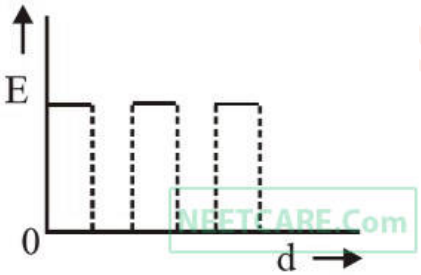
(B)



(C)



(D)



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Q27. AIPMT 2012 Mains

A parallel plate capacitor has a uniform electric field E in the space between the plates. If the distance between the plates is d and area of each plate is A, the energy stored in the capacitor is

(A) $\frac{1}{2}\varepsilon_0 E^2$ (B) $\frac{E^2 Ad}{\varepsilon_0}$

(C) $\frac{1}{2}\varepsilon_0 E^2 Ad$ (D) $\varepsilon_0 EAd$

Open solution

Q28. AIPMT 2011 Prelims

A parallel plate capacitor has a uniform electric field E in the space between the plates. If the distance between the plates is d and area of each plate is A, the energy stored in the capacitor is

(A) $\frac{1}{2}\varepsilon_0 E^2$ (B) $\frac{E^2 Ad}{\varepsilon_0}$

(C) $\frac{1}{2}\varepsilon_0 E^2 Ad$ (D) $\varepsilon_0 EAd$

Open solution

Q29. AIPMT 2010 Mains

Two parallel metal plates having charges +Q and $-$Q face each other at a certain distance between them. If the plates are now dipped in kerosene oil tank, the electric field between the plates will

(A) become zero (B) increase (C) decrease (D) remain same

Open solution

Q30. AIPMT 2010 Prelims

A series combination of n_1 capacitors, each of value C_1, is charged by a source of potential difference 4V. When another parallel combination of n_2 capacitors, each of value C_2, is charged by a source of potential difference V, it has the same (total) energy stored in it. as the first combination has. The value of C_2. in terms of C_1, is then

(A) $\frac{2C_1}{n_1 n_2}$ (B) $16\frac{n_2}{n_1}C_1$ (C) $2\frac{n_2}{n_1}C_1$ (D) $\frac{16C_1}{n_1 n_2}$

Open solution

Q31. AIPMT 2009

Three capacitors each of capacitance C and of breakdown voltage V are joined in series. The capacitance and breakdown voltages of the combination will be

(A) $3C, \frac{V}{3}$ (B) $\frac{C}{3}, 3V$ (C) $3C, 3V$ (D) $\frac{C}{3}, \frac{V}{3}$

Open solution

Q32. AIPMT 2008

A parallel plate capacitor has a uniform electric field E in the space between the plates. If the distance between the plates is d and area of each plate is A, the energy stored in the capacitor is

(A) $\frac{1}{2}\varepsilon_0 E^2$ (B) $\frac{E^2 Ad}{\varepsilon_0}$

(C) $\frac{1}{2}\varepsilon_0 E^2 Ad$ (D) $\varepsilon_0 EAd$

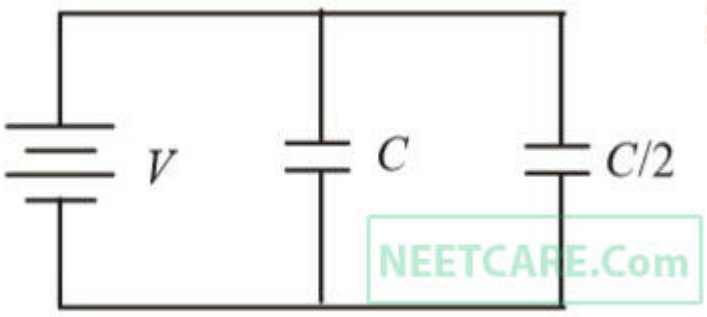
Open solution

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Q33. AIPMT 2007

Two condensers, one of capacity C and other of capacity C/2 are connected to a V-volt battery, as shown in the figure. The work done in charging fully both the condensers is

Open solution



(A) $\frac{1}{4}CV^2$ (B) $\frac{3}{4}CV^2$ (C) $\frac{1}{2}CV^2$ (D) $2CV^2$

Q34. AIPMT 2006

A parallel plate air capacitor is charged to a potential difference of V volts. After disconnecting the charging battery the distance between the plates of the capacitor is increased using an insulting handle. As a result the potential difference between the plates

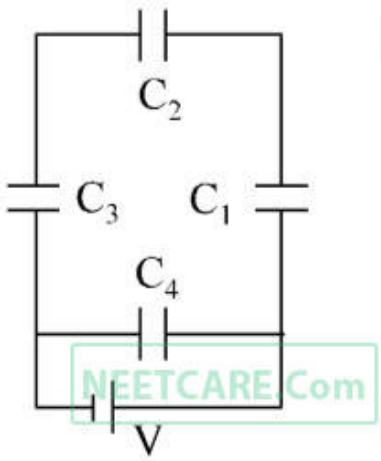
Open solution

(A) increases (B) decreases (C) does not charge (D) becomes zero

Q35. AIPMT 2005

A network of four capacitors of capacity equal to C_1 = C, C_2 = 2C, C_3 = 3C and C_4 = 4C are connected to a battery as shown in the figure. The ratio of the charges on C_2 and C_4 is

Open solution



(A) 4/7 (B) 3/22 (C) 7/4 (D) 22/3

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Q36. AIPMT 2003

Three capacitors each of capacity 4 μF are to be connected in such a way that the effective capacitance is $6\mu F$. This can be done by

(A) connecting all of them in series

(B) connecting them in parallel

(C) connecting two in series and one in parallel

(D) connecting two in parallel and one in series

Open solution

Q37. AIPMT 2002

A capacitor of capacity C_1 charged upto V volt and then connected to an uncharged capacitor of capacity C_2. The final potential difference across each will be

(A) $\frac{C_1V}{C_1+C_2}$

(B) $\frac{C_2V}{C_1+C_2}$

(C) $\left(1-\frac{C_2}{C_1}\right)V$

(D) $\left(1+\frac{C_2}{C_1}\right)$

Open solution

Q38. AIPMT 2001

Energy per unit volume for a capacitor having area A and separation d kept at potential difference V is given by

(A) $\frac{1}{2}\varepsilon_0\frac{V^2}{d^2}$

(B) $\frac{1}{2\varepsilon_0}\frac{V^2}{d^2}$

(C) $\frac{1}{2}CV^2$

(D) $\frac{Q^2}{2C}$.

Open solution

Q39. AIPMT 2000

A capacitor is charged with a battery and energy stored is U. After disconnecting battery another capacitor of same capacity is connected in parallel to the first capacitor. Then energy stored in each capacitor is

(A) $U/2$ (B) U/4 (C) 4U (D) 2U

Open solution

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Answer Key

1. **A**	2. **B**	3. **A**	4. **C**	5. **A**
6. **B**	7. **C**	8. **A**	9. **B**	10. **D**
11. **B**	12. **D**	13. **A**	14. **D**	15. **C**
16. **D**	17. **C**	18. **B**	19. **C**	20. **A**
21. **A**	22. **C**	23. **B**	24. **D**	25. **B**
26. **C**	27. **C**	28. **C**	29. **C**	30. **D**
31. **B**	32. **C**	33. **B**	34. **A**	35. **B**
36. **C**	37. **A**	38. **A**	39. **B**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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